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TAMPINES JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME	

CIVICS GROUP	1	2				TUTOR NAME	

Biology **8875/01**

Paper 1 Multiple Choice

Tuesday, 24 September 2013

1 hour

Additional Materials: Optical Answer Sheet (OAS)

READ THESE INSTRUCTIONS FIRST

Write your name and Civics Group in the spaces at the top of this page and on any separate writing paper used.

There are **30** questions in this paper.

Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate **OAS.**

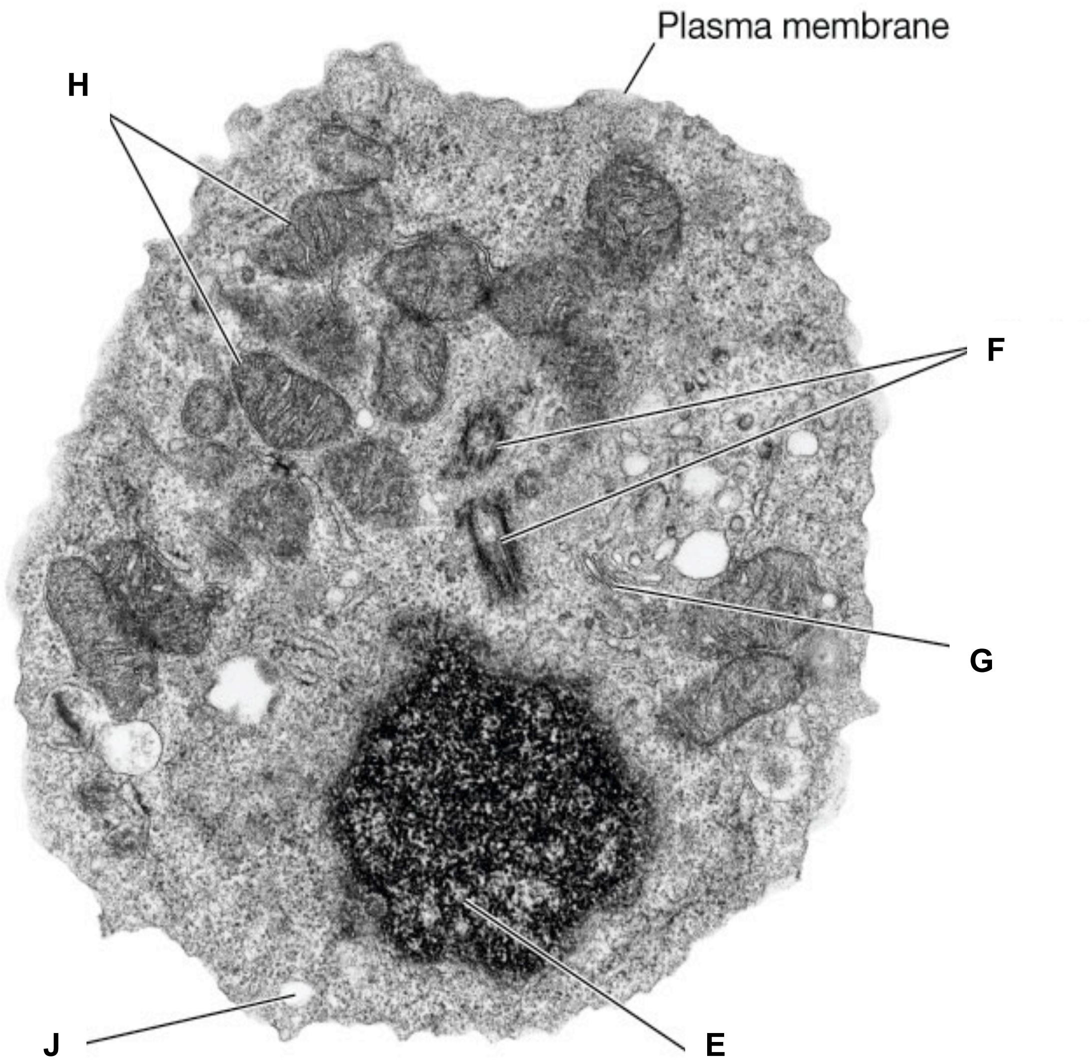
Read the instructions on the OAS very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

MULTIPLE CHOICE QUESTIONS (30 MARKS)

1. An electron micrograph of a cell is shown below.



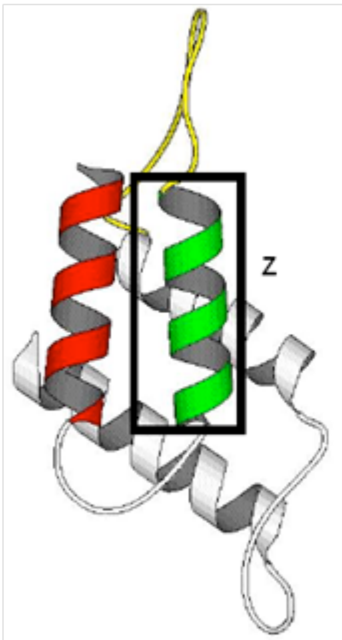
Match the organelles **E, F, G, H** and **J** associated with the cellular processes listed.

	E	F	G	H	J
A	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation	Packaging of secretory products
B	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication	Packaging of secretory products
C	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products	DNA replication
D	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation	Digestion of material

2. Which row correctly links a feature of a phospholipid molecule with its function in membranes?

	Both polar and non-polar (amphipatic)	Phosphate group	Unsaturated fatty acids	Saturated fatty acids
A	Allowing diffusion of small molecules	Aiding membrane stability	Decreasing fluidity	Binding proteins
B	Forming a bilayer	Repelling charged molecules	Increasing fluidity	Decreasing fluidity
C	Forming glycolipids	Binding proteins	Increasing permeability	Increasing fluidity
D	Forming vesicles	Forming glycolipids	Binding proteins	Decreasing permeability

3. The diagram below shows the structure of a general transcription factor.



Which correctly describes the structure of the part enclosed in box Z?

	Number of Amino Acids Present	Types of bonds Present
A	Cannot be determined	Hydrogen Bonds only
B	Cannot be determined	Hydrogen Bonds and hydrophilic bonds
C	Approximately 11 amino acids	Hydrogen bonds only
D	Approximately 11 amino acids	Hydrogen bonds and hydrophobic bonds

4. Each polypeptide chain of haemoglobin contains a number of regions in the form of an alpha helix.

Which feature of the haemoglobin molecule is responsible for this?

A bonding between the four polypeptide chains

B hydrophobic interactions at the centre of the molecule

C the amino acid sequence of the globin

D the presence of iron in the haem groups

5. Which of the following statement about enzymes is inaccurate?

A Enzyme function is not influenced by factors such as lead nitrate.

B Enzyme function is reduced if the three-dimensional structure or conformation of an enzyme is altered.

C Enzymes increase the rate of chemical reaction by lowering activation energy barriers.

D Enzymes may require a non-protein cofactor or ion for catalysis to take place.

6. Which of the following does not describe the active site of an enzyme?

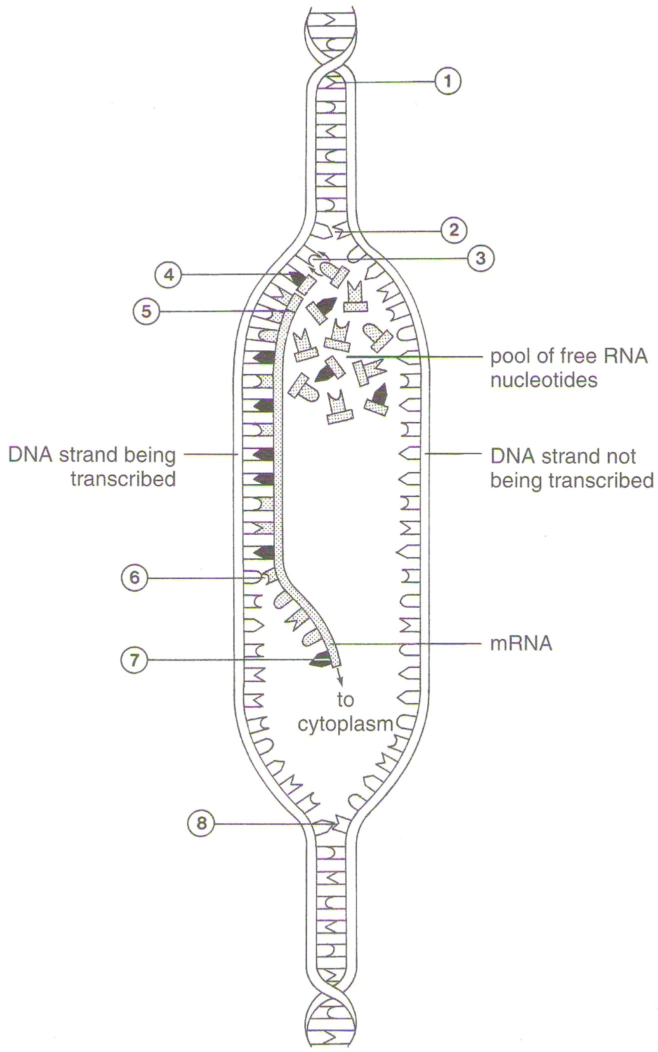
A Its specificity is defined by the arrangement of the monomers.

B It is usually a crevice or cleft.

C It initially binds substrates by weak attractions.

D It is two-dimensional in structure.

7.



Which of the following statements is true?

A The bond in 1 is phosphodiester bond.

B The bond in 5 is peptide bond.

C 7 refers to the 5' end of the mRNA.

D 8 is showing unwinding of the DNA molecule.

8. The following coding sequence of DNA is taken randomly from a bacterial genome.

3’ TTACGCTTCGAAATAGGAATATCATAGGCT 5’

This sequence is cloned into a plasmid and transformed into a suitable host. What would be the first four amino acids of a polypeptide generated from this sequence as expressed by the host?

The mRNA codons for some amino acids are shown in the table below:

Arg	CGA, CGG, AGA, AGG	Leu	CUU, CUC, CUA, CUG
Asp	GAU, GAC	Lys	AAA, AAG
Ile	AUU, AUC, AUA	Phe	UUU, UUC
Start	AUG	Ser	UCA, UCG, AGU, AGC
Stop	UAG, UGA, UAA	Tyr	UAU, UAC

A Met-Arg-Ser-Phe

B Met-Arg-Ser-Lys

C Met-Ile-Phe-Leu

D Met-Tyr-Lys-Asp.

9. What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversion of segments of chromosomes do occur.
C	Both affect DNA sequence	Chromosomal aberrations may occur across chromosomes but gene mutations occur within a chromosome.
D	Both will not result in a different in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

10. Which of the following is unique to mitosis and not a part of meiosis?

A Homologous chromosomes pair forming bivalent.

B Homologous chromosomes cross over.

C Chromatids are separated during anaphase.

D Homologous chromosomes behave independently.

11. Cells treated with colchicine often mutate. They fail to produce spindles during cell division. What would be the result of this?

A chromosome deletion

B chromosome translocation

C gene deletion

D polyploidy

12. Albinism in humans is controlled by a recessive allele. How many copies of this allele will be found at one of the poles of a cell at telophase I in an albino person?

A 23

B 4

C 2

D 1

13. Skin colour in a variety of pumpkins is controlled by two pairs of alleles, Pp and Rr, which segregate independently.

The allele P is dominant and must be present for the development of pigmentation in the skin. Pumpkins without pigment are white. Allele R is dominant and produces a red pigment in the skin. The recessive allele r gives a yellow colour.

In a testcross with a parent of unknown genotype, a phenotypic ratio of 1 red : 1 white was obtained.

What was the genotype of the unknown parent?

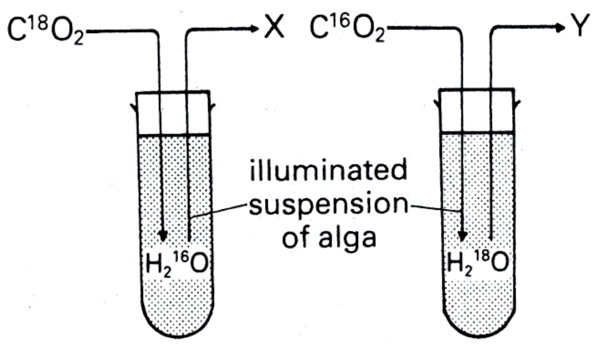
A PPRR

B PpRR

C PPRr

D Pprr

14. In the experiment shown in the diagram, the oxygen given off at **X** and **Y** will be labelled with the isotopes _____ and _____ respectively.



	X	Y
A	^{16}O	^{16}O
B	^{16}O	^{18}O
C	^{18}O	^{16}O
D	^{18}O	^{18}O

15. The weed killer DCMU blocks the flow of electrons from the electron transport chains in photophosphorylation.

Why does this kill the plant?

A Active transport of mineral ions is prevented.

B ATP and reduced NADP are not produced.

C Photoactivation of the chlorophyll cannot occur.

D Photolysis of water does not occur.

16. In a classic experiment on photosynthesis, biochemist Robert Hill from the Department of Biochemistry at Cambridge demonstrated that an illuminated *in vitro* suspension of isolated chloroplasts could produce oxygen in the presence of a hydrogen acceptor such as methylene blue. In this case, methylene blue is reduced. Which of the following compound replaces methylene blue in the intact photosynthesising plant?

A Adenosine triphosphate

B Carbon dioxide

C Nicotinamide adenine dinucleotide phosphate

D Nicotinamide adenine dinucleotide

17. To what extent can glucose be oxidised by the soluble portion of cytoplasm of muscle cells, from which all organelles have been removed?

A Acetyl-CoA but no further

B Carbon dioxide and water

C Lactate but no further

D Pyruvate but no further

18. In which of the following changes is most energy released during the metabolism of carbohydrates?

A Maltose to glucose

B Glucose to phosphoglyceraldehyde

C Phosphoglyceraldehyde to pyruvic acid

D Pyruvic acid to carbon dioxide and water

19. Which molecule is common to both respiration and photosynthesis?

A Hexose phosphate

B NADP

C Pentose phosphate

D Triose phosphate

20. Which of the following increases the number of different alleles in a population?

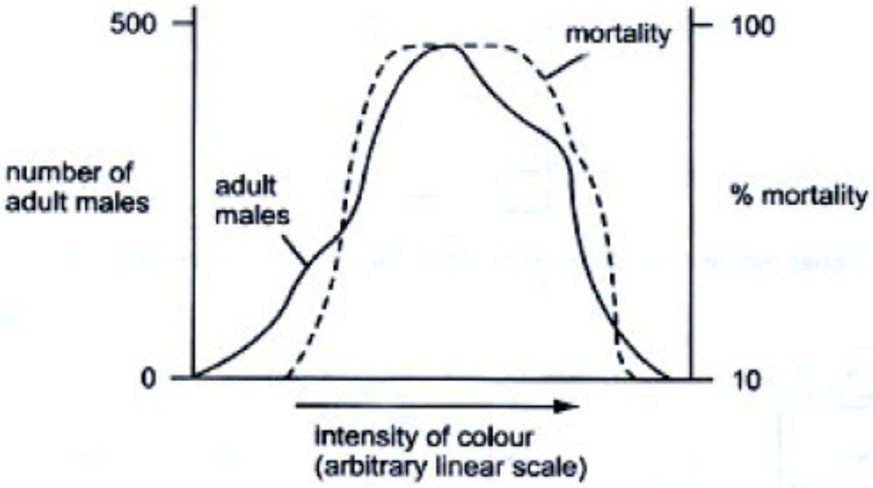
A crossing over

B gene mutation

C random fusion of gametes

D random assortment of chromosomes in meiosis

21. The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity. Which conclusion can be made from this graph?



A Colour variation is environmentally induced.

B Colour variation is genetically determined.

C Extreme forms are favoured by natural selection.

D The species shows discontinuous variation with respect to colour.

22. Approximately 1 in 20 Europeans are heterozygous for a recessive allele responsible for the genetic condition, cystic fibrosis (CF). People who are homozygous for CF have a reduced life expectancy. Heterozygotes are more resistant to some bacterial infections of the gut, such as typhoid fever, than homozygotes for the normal, dominant allele.

What could explain the high incidence of the recessive CF allele in the European population?

A natural selection favouring heterozygotes

B natural selection favouring homozygotes for the recessive CF allele

C lack of genetic drift in the European population

D Mutation

23. What is one similarity and one difference between classification and phylogeny?

A Both organise organisms into groups based on their characteristics but phylogeny takes into consideration the evolutionary relationships between groups.

B Both put organisms together according to increasingly inclusive groups but classification does not show a branching pattern of evolutionary relationships.

C Both rely on physical characteristics of organisms but phylogeny does not require scientific names.

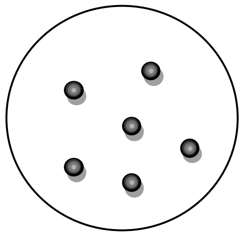
D Both take into account the common ancestors of organisms but classification is hierachical in nature.

24. Which uses of the information from the human genome project are generally considered to be unethical?

1. an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases
2. genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships
3. cytologists developing tests for only some defective genes
4. doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes
5. genetic counselors giving specific lifestyle information only to people genetically predisposed to risks
6. parents choosing embryos for implantation only after ante-natal tests for acceptable genes

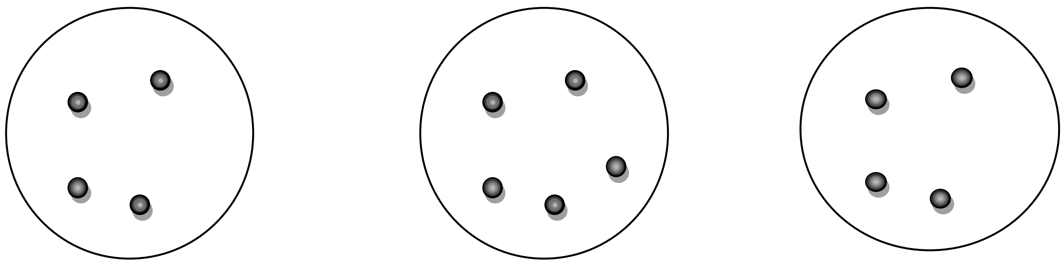
A	1 and 3
B	1 and 6
C	2 and 5
D	3 and 4

25. The insulin gene and a plasmid with ampicilin and tetracycline resistant genes were cut using the same restriction enzyme, *Eco*RI. The cut plasmid and insulin gene was mixed together and DNA ligase was allowed to react. Subsequently the mixture was heat shocked at 42 degrees with *E. coli*. The mixture was then plated on nutrient agar to obtain a master plate with colonies shown below.



Master plate (nutrient agar)

Three replica plates containing different antibiotics were obtained from the master plate. The diagram below shows the results of the replica plates.



Nutrient agar with ampicillin

Nutrient agar with tetracycline

Nutrient agar with tetracycline and ampicillin

Which of the following can be concluded from the results obtained?

A	The ampicilin resistant gene is inactivated by insertion of the insulin gene
B	The tetracycline resistant gene is inactivated by insertion of the insulin gene.
C	The tetracycline resistant gene does not act as a selectable marker
D	The ampicillin resistant gene does not act as a selectable marker.

26. A student performed an experiment to determine if a particular gene has been inserted in a genetically modified organism.

i. Transfer of DNA to nitrocellulose membrane
ii. Restriction digestion of genomic DNA
iii. Cleaved DNA separated using gel electrophoresis
iv. Create radioactive probe
v. Incubate probe and membrane

Which is the correct sequence to the above steps?

A	ii → iii → i → v → iv.
B	ii → iii → i → iv → v.
C	iv → v → i → ii → iii.
D	iii → i → ii → iv → v.

27. The genetically engineered super salmon was created from Atlantic salmon stocks and are capable of growing to a large size in 14 months. Which of the following is not a benefit intended from the crop?

A Higher yield for farmers

B Minimising pollution

C Decreasing the food consumption of the crops in their lifetime

D Increase in supply to meet world's demand

28.

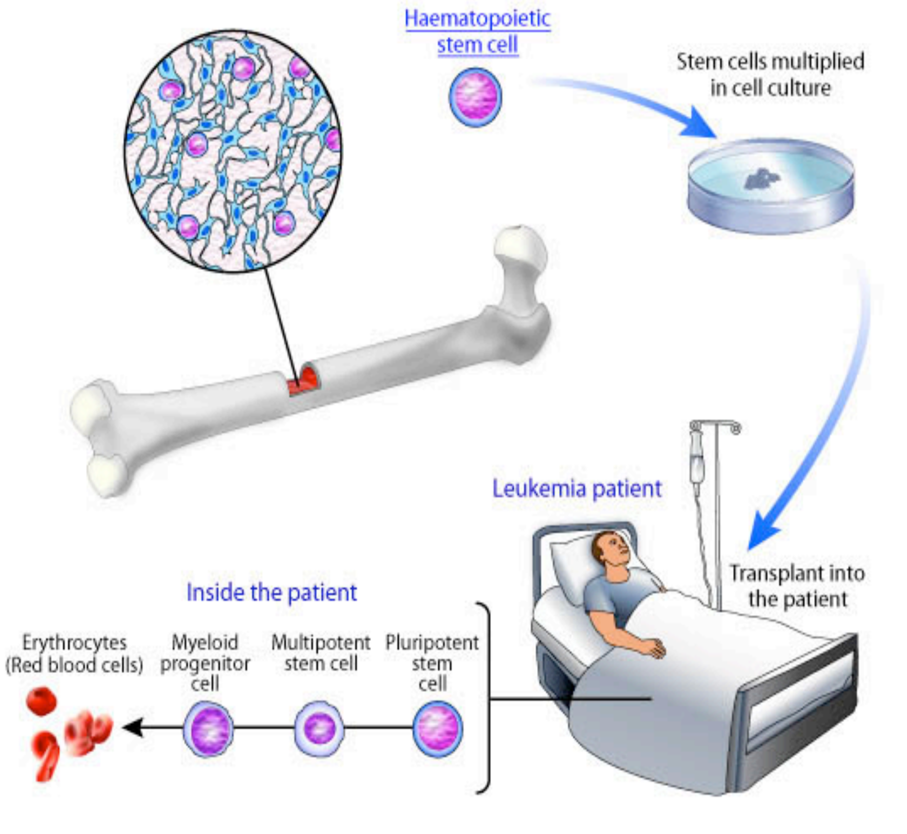


Illustration by Cell Imaging Core of the Center for Reproductive Sciences.

Which of the following options is not a characteristic of the form of treatment shown above?

1 Easy to extract in laboratory
2 Readily available
3 Difficult to control
4 Limited longevity

A 1 only **B** 1 and 2 **C** 1 and 3 **D** 1, 2, 3 and 4

29. Germ-line gene therapy has not been prevalent in medical treatment because of many reasons. Which of the following is/are implications of germ-line therapy?

1 unforeseeable clinical risks
2 opens problems for social discrimination
3 cost
4 issue of consent

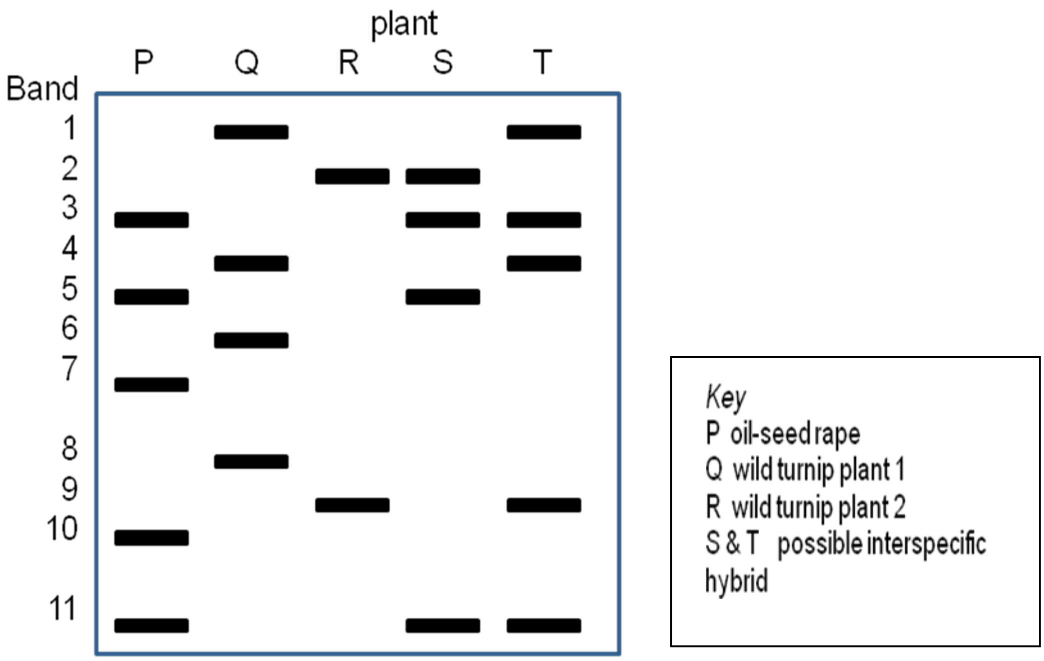
A 1 only **B** 2, 3 and 4 **C** 1, 2 and 4 **D** 1, 2, 3 and 4

30. A possible route of escape of an inserted gene from a genetically engineered crop of oil-seed rape, *Brassica napus*, is into populations of the wild turnip, *Brassica rapa*.
Two populations of wild turnip growing next to large fields of a cultivar of oil-seed rape were studied. Seeds were collected from these wild turnips plants and their DNA analysed using restriction enzymes and electrophoresis.

Using the key provided below, determine

(i) whether plants S and/or T are interspecific hybrids and

(ii) the parental plants they are derived from



	Interspecific hybrid	*Parent plants*
A	Both S and T	S: P x R T: P x Q
B	Both S and T	S: P x Q T: P x R
C	T only	Q x R
D	S only	P x R